

# IDEATION PHASE

## Brainstorm & Idea Prioritization

**Project Name:** AI BlogNest API

**Team ID:** SWTID-2026-6999

**Date:** 20 September 2026

### Step 1: Team Gathering, Collaboration and Selection of the Problem Statement

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#### Team Gathering and Collaboration

The team discussed common problems related to blog content creation and management. The discussion focused on difficulties in creating blog content, organizing posts using categories and tags, managing comments, controlling user permissions, and searching for relevant blog content.

Different possible solutions were discussed by considering the use of Artificial Intelligence, secure backend technologies, and intelligent search techniques. After discussing the different problems and possible solutions, the team selected a problem related to efficient and intelligent blog content management.

#### Problem Statement

Content creators and blogging platforms face difficulties in efficiently creating and managing blog content, organizing posts, controlling user access, moderating comments, and finding relevant content through traditional search methods.

#### Need for the Solution

A system is required to reduce the manual effort involved in content creation and summarization, provide secure access to different users, organize blog content effectively, and improve the process of searching and discovering relevant articles.

#### Selected Problem Statement

*How can an AI-powered backend system be developed to simplify blog content creation and management while providing secure role-based access and intelligent search capabilities?*

### Step 2: Brainstorm, Idea Listing and Grouping

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The team brainstormed a wide range of ideas relevant to the problem statement and grouped them under the following functional categories:

| Group                    | Ideas                                                                                                                                                                                                      |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AI Content Creation      | <ul style="list-style-type: none"> <li>AI Blog Generation</li> <li>AI Blog Summarization</li> <li>AI-generated Blog Titles</li> <li>Automatic Tag Recommendation</li> </ul>                                |
| Blog Management          | <ul style="list-style-type: none"> <li>Blog CRUD Operations</li> <li>Draft Management</li> <li>Publishing Status Management</li> <li>Category Management</li> <li>Tag-based Organization</li> </ul>        |
| Search and Discovery     | <ul style="list-style-type: none"> <li>Keyword Search</li> <li>MongoDB Atlas Search</li> <li>Semantic Search</li> <li>Vector Search</li> <li>Embedding Generation</li> </ul>                               |
| User Access and Security | <ul style="list-style-type: none"> <li>User Registration</li> <li>User Login</li> <li>JWT Authentication</li> <li>Role-Based Access Control</li> <li>Password Hashing</li> <li>Input Validation</li> </ul> |
| Comments and Moderation  | <ul style="list-style-type: none"> <li>Add Comments</li> <li>View Comments</li> <li>Edit Comments</li> <li>Delete Comments</li> <li>Comment Moderation</li> </ul>                                          |
| Backend Architecture     | <ul style="list-style-type: none"> <li>REST APIs</li> <li>MVC Architecture</li> <li>Service Layer</li> <li>MongoDB Database</li> <li>Postman API Testing</li> </ul>                                        |

### Step 3: Idea Prioritization

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The ideas were prioritized based on their importance and feasibility, as summarized below.

#### High Importance and High Feasibility

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| <ul style="list-style-type: none"> <li>User Registration and Login</li> <li>JWT Authentication</li> <li>Role-Based Access Control</li> </ul> |
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- Blog CRUD
- Category Management
- Comment Management
- AI Blog Generation
- AI Blog Summarization

### **High Importance and Moderate Feasibility**

- MongoDB Atlas Search
- MongoDB Atlas Vector Search
- Semantic Search
- Embedding Generation

### **Moderate Importance and High Feasibility**

- Blog Filtering
- Tag Management
- Pagination
- Comment Status Management
- API Testing

### **Future Enhancement Ideas**

- Advanced Analytics
- Personalized Recommendations
- Frontend Dashboard
- AI-based SEO Assistance
- Multi-language Content Generation

## Final Selected Idea

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### AI BlogNest API

AI BlogNest API is selected as the final project idea because it combines:

- Secure user authentication
- Role-Based Access Control
- Blog management
- Categories and comments
- AI-powered blog generation
- AI-powered summarization
- MongoDB Atlas Search
- MongoDB Atlas Vector Search
- Semantic Search

*The final solution is a backend REST API that combines blog content management with Artificial Intelligence and intelligent search capabilities.*